

A Recurrent Algorithm for Conway Matrices Related to a Non-transitive Head-or-Tail Game

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Our paper deals with the well-known problem of binary string overlapping in connection with the so-called “non-transitive head-or-tail game,” known also as “Penney Ante game” (cf. [3, 4]). We suggest a recurrent algorithm to calculate Conway matrices related to this game. The procedure is implemented in *Mathematica*. Visualization of these matrices illustrates the choices of the winning strategy. Though the problem was formulated more than 30 years ago, and the solution is known, it still attracts attention, and different approaches to dealing with it are suggested (see, e.g., [1, 6]; for the strategy of the first player see [2]). The game looks simple, but there are speculations that it could be applied (with proper modifications) to real-life situations, e.g. to the strategy of stock market investments [1]. Since the problem is described in a classical textbook [7, section 8.4] (the case of two players; the more general case is described in [4]), we give only a brief description here:

Two players, (1) and (2), agree on some integer $n \geq 2$. Then both of them select a binary n -word (“head”=0, “tail”=1), say w_1 and w_2 , and begin flipping a coin until either w_1 or w_2 appears as a block of n consecutive outcomes. Player (1) wins if w_1 appears before w_2 does. The problem is to find the probability $P_n(w_1, w_2)$ that player (1) will win for the chosen w_1 and w_2 .

A solution to the problem was proposed by J.H. Conway (but was not published, cf. [2]) and the key tool of his solution is the so-called Conway matrix $C_n = (c_{ij})$, $0 \leq i, j \leq 2^n - 1$, where indices i, j encode n consecutive outcomes as a binary word in the usual way : HTTHT $\leftrightarrow$ 01101 $\leftrightarrow$ 13.

With matrix C_n known, one can easily find the matrix P_n of the required probabilities by the Conway formula

$$(P_n)_{ij} = \frac{c_{jj} - c_{ji}}{(c_{ii} + c_{jj}) - (c_{ij} + c_{ji})}, \quad i \neq j; \quad (P_n)_{ii} = 0. \quad (1)$$

Here we suggest a recurrent algorithm to calculate Conway matrices $C_n = (c_{ij})$.

Computer algebra systems, such as *Mathematica* or *Maple*, provide good facilities for symbolic and numeric calculations making bulky calculations easy. They also allow quick illustrative examples and tests of hypotheses, and have convenient graphics tools. We used *Mathematica* in this study.

We describe the matrix $C_n = (c_{ij})$ below. It turns out that the structure of the Hadamard transform $\hat{C}_n$ of the matrix C_n is much simpler than that of the original matrix C_n , and this allows us to construct a simple and effective recurrent algorithm for calculation and visualization of matrices C_n and the corresponding matrices P_n .

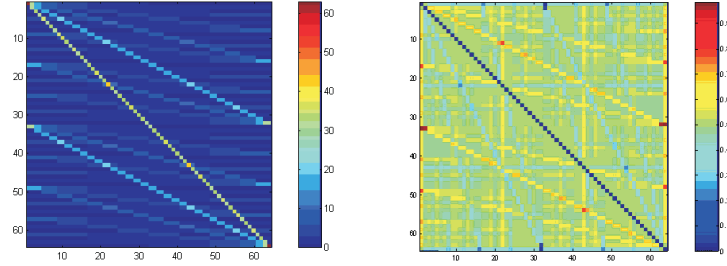


Figure 1: Conway matrix C_6 (left) and matrix of probabilities P_6 (right).

Let V_n be the space of the binary n -words, i.e. the n -dimensional vector space over the field $Z_2 = \{0, 1\}$, and k an integer, $0 \leq k \leq n$. We say that binary vector $u = (\varepsilon_0, \varepsilon_1, \dots, \varepsilon_{n-1})$ k -overlaps vector $v = (\eta_0, \eta_1, \dots, \eta_{n-1})$ if $(\varepsilon_k, \varepsilon_{k+1}, \dots, \varepsilon_{n-1}) = (\eta_0, \eta_1, \dots, \eta_{n-k-1})$ and write this as $u =_k v$. The Conway number of the pair (u, v) is

$$c(u, v) = \sum_{k=0}^{n-1} 2^{n-1-k} \|u =_k v\|, \quad (2)$$

where $\|u =_k v\|$ equals 1 (resp. 0) if $u =_k v$ is “true” (resp. “false”).

For every n the set of all Conway numbers constitutes the $2^n \times 2^n$ Conway matrix $C_n = (c(u, v))$ with the natural ordering of binary vectors as binary expansions of the indexes $0, 1, \dots, 2^n - 1$.

The algorithm was implemented in Mathematica as a program CONWAY.nb.

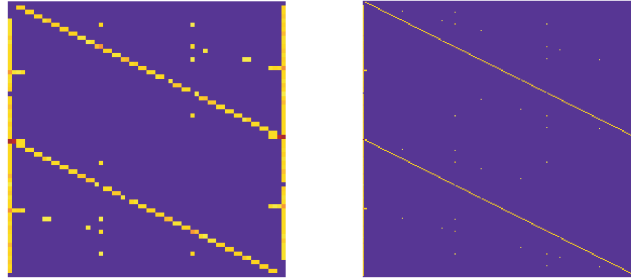


Figure 2: Matrices of probabilities P_6 (left) and P_8 (right) with elements less than 0.6 truncated.

Figure 1 visualizes results of calculations with CONWAY[6,0] for matrices C_6 and P_6 , whereas Figure 2 (left) demonstrates the same matrix P_6 with all the elements which are less than 0.6 truncated. They are shown in dark blue. Truncation at 0.6 means that the probability that “row wins the column” is not less than 60% (blue background corresponds to probabilities less than 60%). As

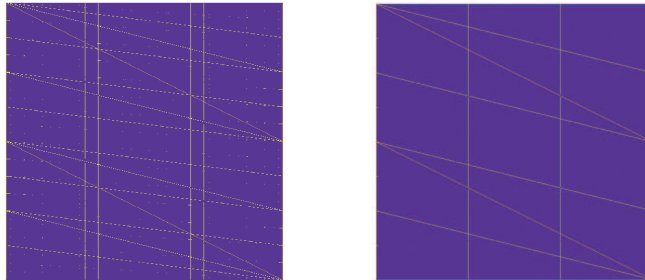


Figure 3: Matrix of probabilities P_{10} . Truncated are elements less than 0.52 (left) and 0.55 (right).

can be seen, whatever word is selected by the “column”, “row” always has an adequate answer (with probability $p \geq 0.6$).

Finally we draw the reader’s attention to the changes in the structure of the truncated probability matrices P_n (with values truncated below some threshold) that occur when the threshold value changes (see Figure 3). It also illustrates the complexity of the problem.

References

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